

Wine Quality Drivers

A Multi-Method Analytical Framework for Physicochemical and Regional Review
Patterns

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Professional Analytical Report

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Executive Summary

Wine quality is influenced by a complex interaction of physicochemical characteristics that affect flavor, aroma, structure, and overall consumer perception. This analysis examines a combined dataset of 6,497 red and white wine samples containing laboratory-measured physicochemical attributes and corresponding quality ratings.

Using a structured Exploratory Data Analysis framework, the report evaluates relationships between chemical characteristics and wine quality, identifies variables with the strongest predictive value, and explores regional review patterns through geographic visualization.

Multiple analytical approaches - exploratory analysis, correlation analysis, logistic regression, Random Forest modeling, and regional review mapping - produced a consistent finding: alcohol content was the most stable positive quality indicator, while volatile acidity and density often showed inverse relationships with quality.

Key conclusion: wine quality is best understood through multiple forms of evidence. Laboratory measurements explain structural quality drivers, while regional review data adds geographic and market-facing context.

Dataset and Methodology

The study uses the Portuguese Vinho Verde Wine Quality dataset, which contains red and white wine observations with 11 numerical predictor variables and a quality rating scale ranging from 3 to 9. The project was organized into five phases: data quality assessment, descriptive statistical analysis, relationship analysis, predictive modeling, and regional review/geographic analysis.

Metric	Value
Observations	6,497
Predictor variables	11
Wine types	Red and white
Target variable	Wine quality
Quality rating scale	3-9
Data type	Structured numerical data

- No missing values were identified.
- No duplicate observations were detected.
- All variables were numerical and suitable for analysis.
- Quality ratings were grouped into Poor, Good, and Excellent categories to support clearer interpretation.

Exploratory Analysis

The exploratory phase established the distribution of quality outcomes and examined how individual physicochemical features relate to wine quality across red and white wines. The majority of observations fell between ratings of 5 and 7, which created a useful baseline for identifying characteristics associated with higher-performing wines.

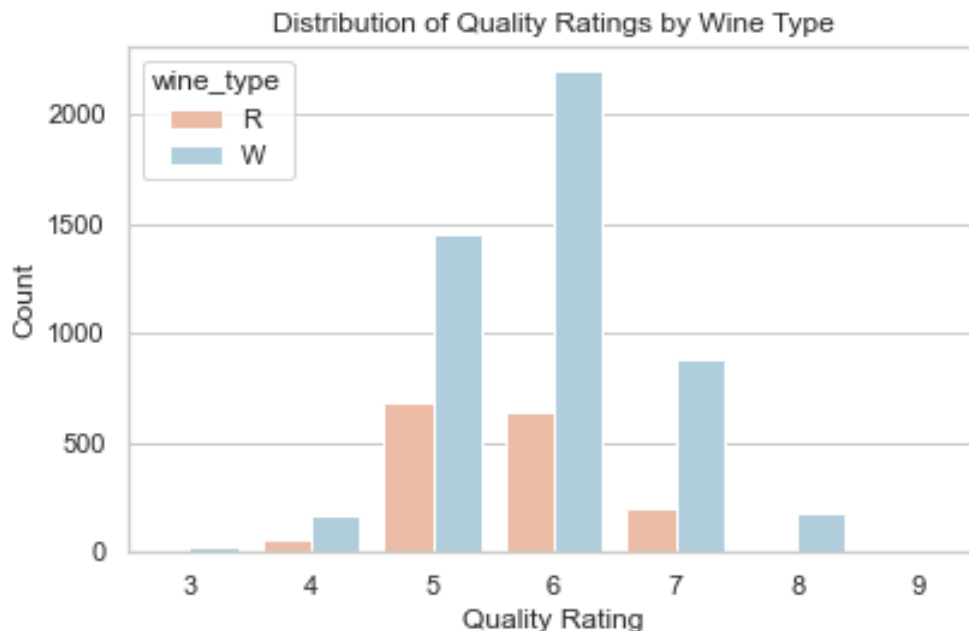


Figure 1. Distribution of quality ratings by wine type. Most wines cluster between quality ratings of 5 and 7, with white wines representing the larger sample share.

The regression views show that alcohol content has one of the clearest positive relationships with wine quality. Volatile acidity shows a negative relationship, meaning wines with higher levels of volatile acidity tend to receive lower quality ratings. Other characteristics, including sulphates, citric acid, density, residual sugar, and sulfur dioxide, show more variable influence by wine type.

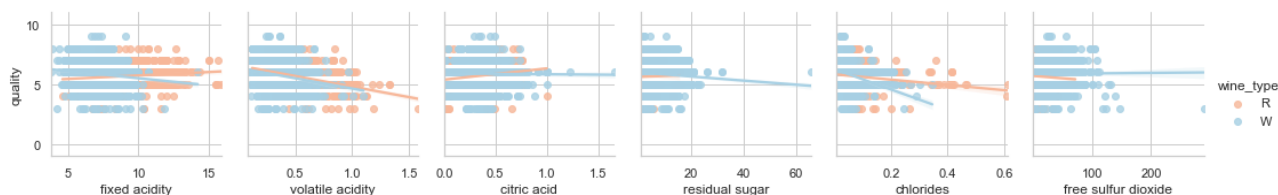


Figure 2. Regression view of selected physicochemical variables and quality by wine type.

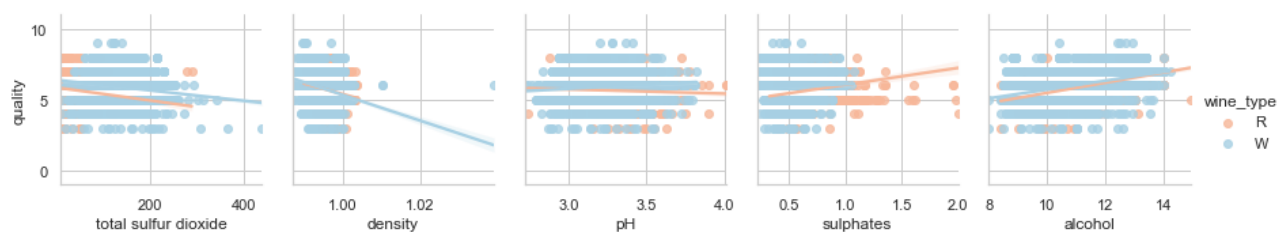


Figure 3. Additional regression view of selected physicochemical variables and quality by wine type.

Correlation Analysis

Correlation analysis provided a quantitative check against the visual relationships observed in the regression plots. Alcohol demonstrated the strongest positive correlation with quality, while density and volatile acidity displayed inverse relationships. This reinforced the value of using both visual exploration and statistical measurement.

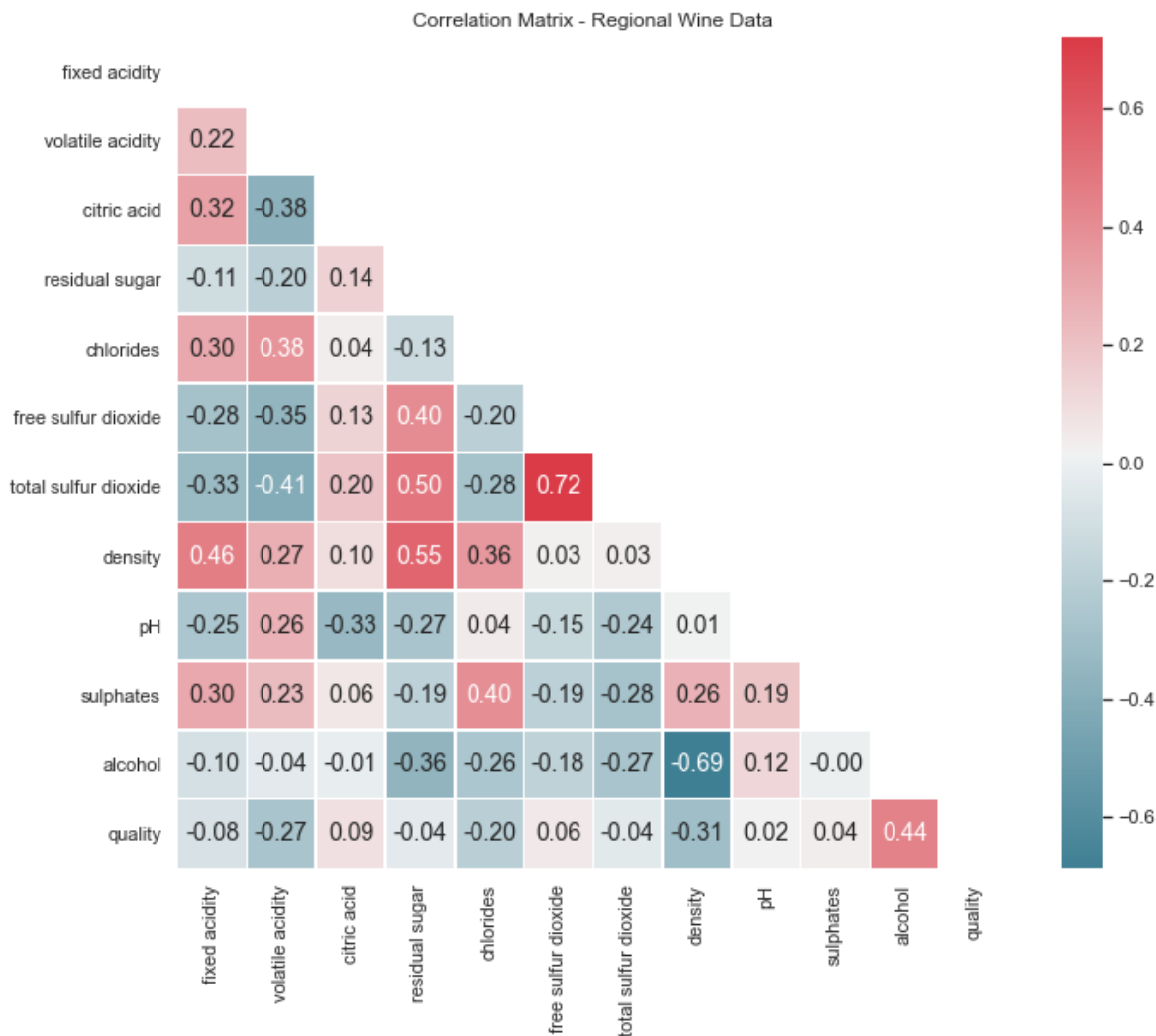


Figure 4. Correlation matrix for the regional wine dataset. Alcohol shows the clearest positive relationship with quality, while density and volatile acidity show negative relationships.

Predictive Modeling

Predictive modeling was used as a validation layer rather than as a stand-alone conclusion. Logistic regression helped identify the direction and relative strength of variables associated with high-quality classification, while Random Forest modeling provided a second view of feature importance.

Across both modeling approaches, alcohol remained the most consistent positive quality indicator. The drivers were not identical across red and white wines, which suggests that quality should not be treated as a universal formula across wine styles.

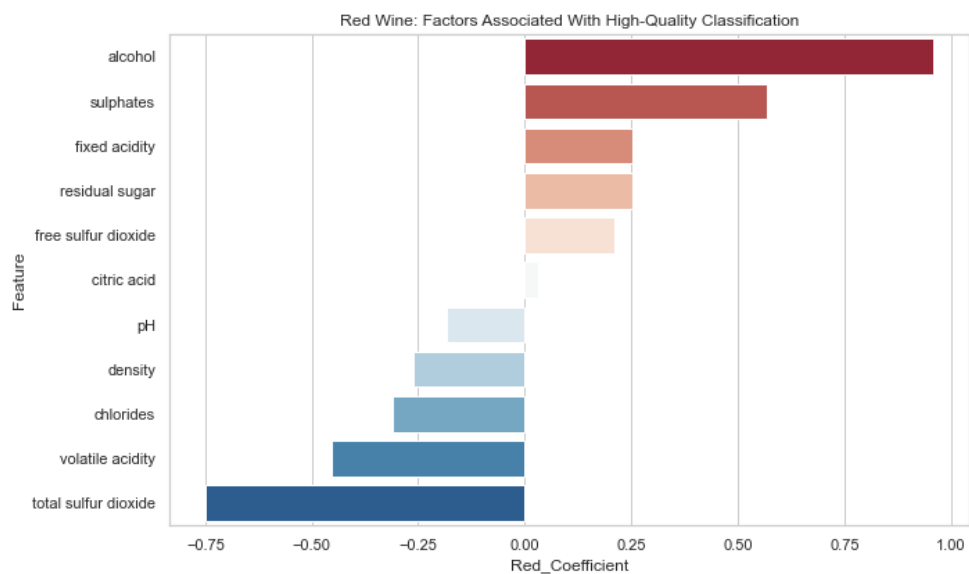


Figure 5. Red wine logistic regression coefficients for high-quality classification.

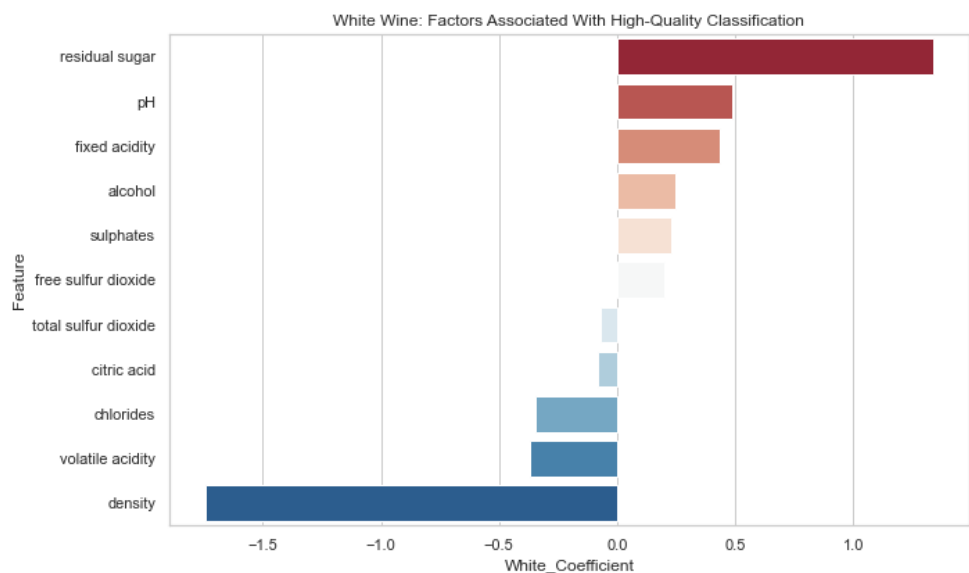


Figure 6. White wine logistic regression coefficients for high-quality classification.

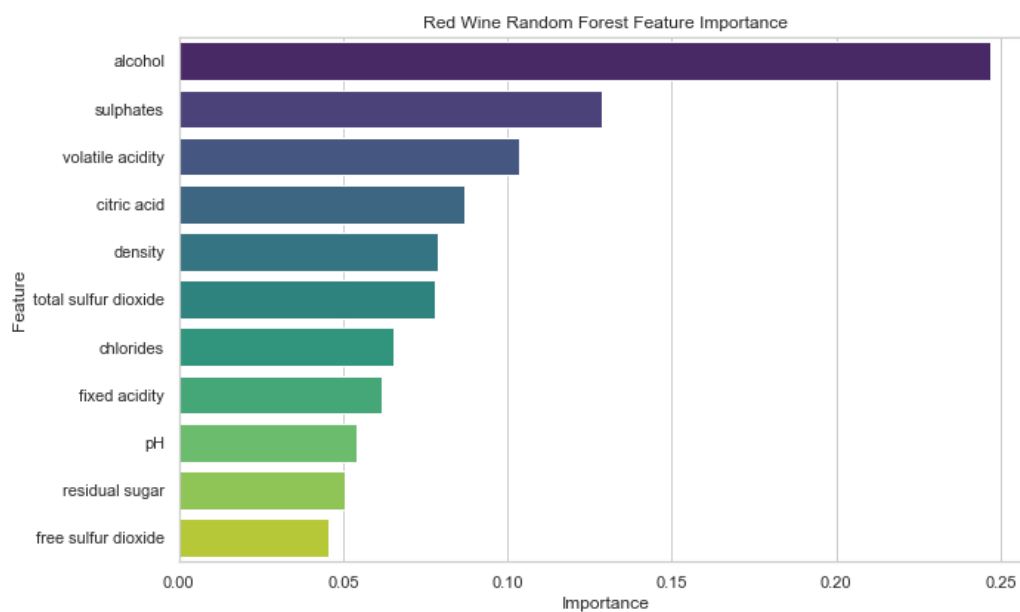


Figure 7. Red wine Random Forest feature importance. Alcohol remains the strongest model feature.

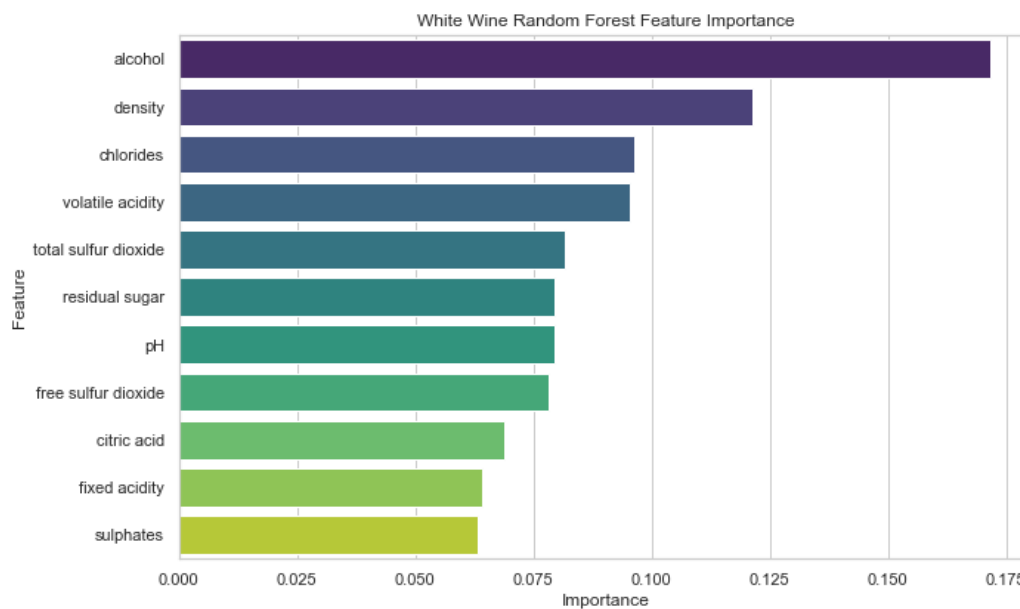


Figure 8. White wine Random Forest feature importance. Alcohol and density are among the leading predictors.

Regional Review Analysis

The second part of the project extended the analysis beyond laboratory measurements by exploring regional review patterns in the Vinho Verde region. This dataset is complementary rather than directly joined to the physicochemical dataset. It provides a market-facing perspective on where highly rated wines appear across subregions.

Subregion performance was evaluated using average review score, review volume, and the proportion of wines receiving scores of 90 points or higher. This allows quality to be viewed through both performance and sample-size context.

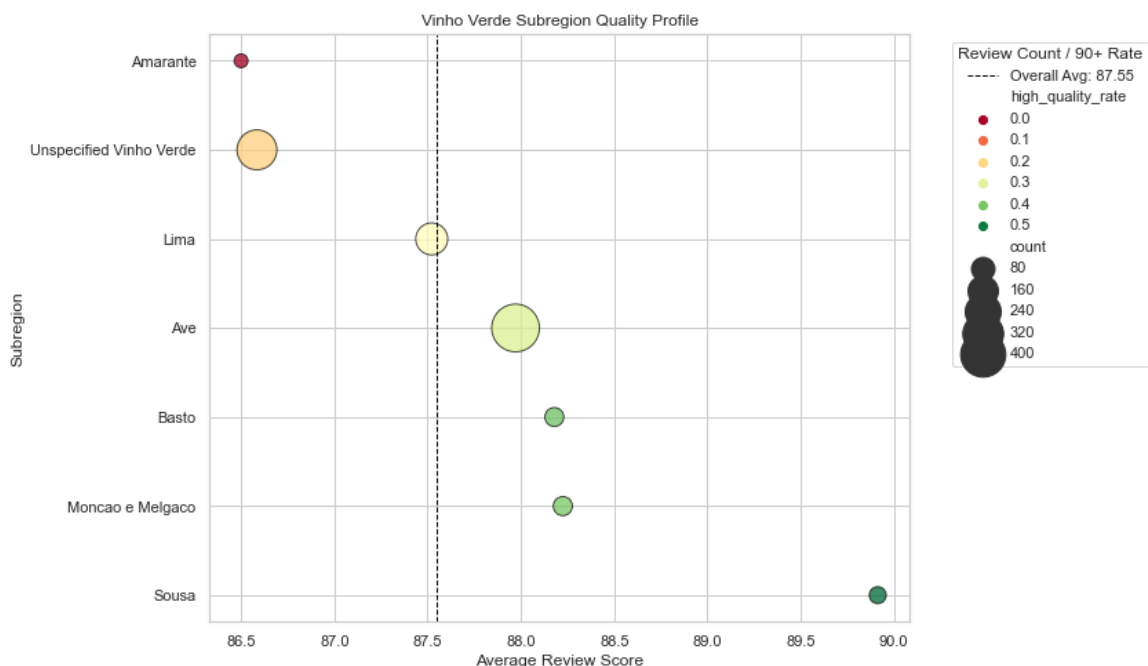


Figure 9. Vinho Verde subregion quality profile comparing average review score and review volume.

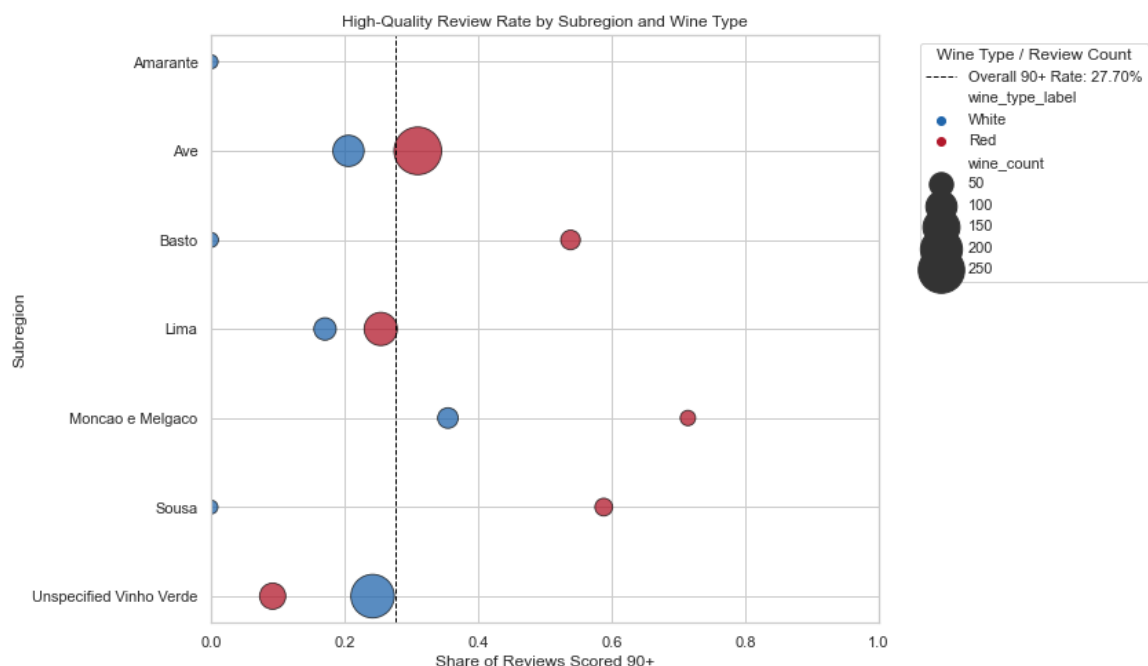


Figure 10. High-quality review rate by subregion and wine type, showing regional variation between red and white wines.

Regional Quality Map

The regional quality map is the strongest geographic visualization in the project. It translates subregional review performance into a spatial view, helping show that high-performing Vinho Verde areas are distributed across multiple parts of the region rather than concentrated in one location.

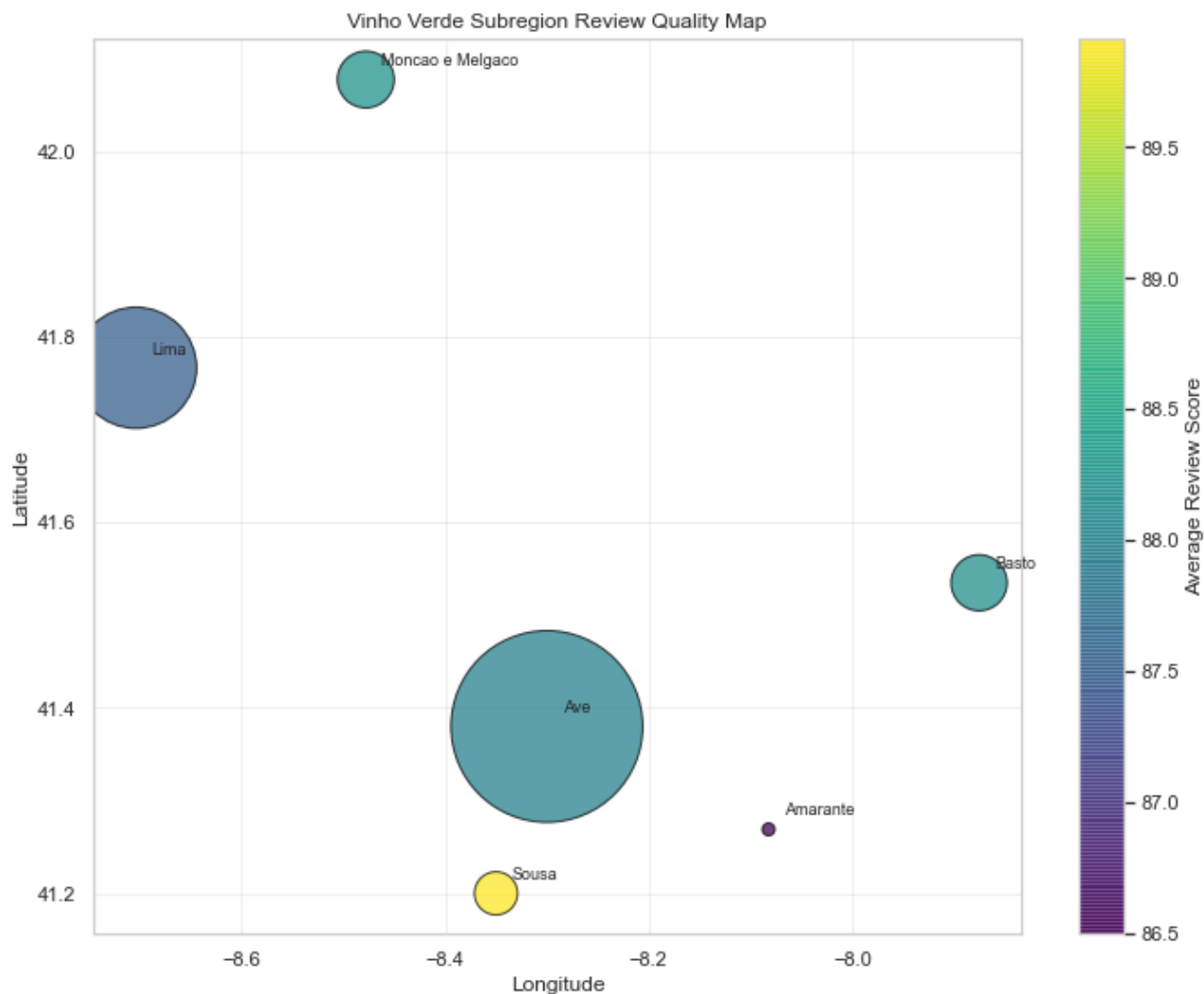


Figure 11. Vinho Verde subregion review quality map. Bubble size reflects review count and color reflects average review score. This is the key regional visualization for the report.

Sousa, Monção e Melgaço, and Basto remain among the strongest-performing subregions based on review quality. Larger subregions such as Ave and Lima contribute greater review volume while maintaining competitive review scores. The static map should be interpreted alongside the statistical analysis rather than as a precise geographic model.

Strategic Insight

This analysis demonstrates that wine quality can be evaluated through two complementary dimensions. The first focuses on measurable physicochemical characteristics that influence wine structure and sensory perception. The second examines how external market validation is distributed across geographic regions.

Key Findings

- Alcohol was the most consistent quality indicator across exploratory analysis, correlation analysis, logistic regression, and Random Forest modeling.
- Quality drivers differ by wine type, suggesting that wine quality cannot be explained using a single universal model.
- Several Vinho Verde subregions outperformed regional averages, indicating that geography may contribute to perceived quality outcomes.
- Chemistry and geography provide complementary perspectives; neither dataset alone fully explains wine quality.

Conclusion

This project examined wine quality through two complementary perspectives: physicochemical characteristics and regional review performance. The laboratory dataset identified measurable characteristics associated with high-quality wines, while the regional review dataset demonstrated that highly rated wines are not distributed evenly across Vinho Verde subregions.

The strongest conclusion is not that a single variable determines wine quality, but that confidence increases when multiple analytical methods point toward similar findings. Alcohol content emerged as the most stable quality indicator, but the interaction of acidity, density, sulfur-related characteristics, wine type, and regional context provides a fuller view of quality outcomes.

Limitations and Future Research

The physicochemical dataset does not contain vineyard, producer, municipality, or subregional information. The regional review dataset does not contain laboratory measurements for the reviewed wines. As a result, the relationship between chemistry and geography cannot be established directly in this project. Future research should develop an integrated wine intelligence dataset that combines laboratory measurements, wine type, producer information, quality ratings, and geographic origin for the same wines.